



Ballistic and mechanical characteristics of paraffin-based solid fuels

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Abstract

In a hybrid rocket, paraffin-based fuels are characterised by high regression rates and zero chemical explosion potential. However, paraffin wax is a brittle material and exhibits poor mechanical strength. The production of large-scale paraffin motors is a challenging task because of the poor structural performance of such motors. The addition of polymers and metallic additives to paraffin wax can considerably enhance mechanical performance. In this study, three different paraffin-based fuels, namely paraffin–aluminium (P–Al), paraffin–boron (P–B), and paraffin–carbon black (P–CB), were prepared using polyethylene (PE) polymer additive. The effects of these additives on the thermal, mechanical, and ballistic performance of the paraffin-based fuels were examined. The mechanical strength of these fuels was evaluated using compressive tests. Differential scanning calorimetry (DSC) experiments were performed to study the thermal decomposition of these paraffin-based fuels. A lab-scale ballistic evaluation motor was used to study the regression rates in a gaseous oxygen environment. The results revealed that the addition of the polymer and metallic additives improved the mechanical strength of fuels. The DSC results demonstrated that the addition of the additives accelerated oxidation and improved the heat release rate during combustion. The oxidation enthalpies for the P–Al, P–B, and P–CB samples were 291, 274, and 268 J/g, respectively. Finally, the ballistic results indicated that the regression rate decreased when 10 wt% PE was added to the paraffin wax, and such depreciation of regression rate due to polymer addition was compensated by the addition of the Al, B, and CB additives.

Keywords Regression rate · Paraffin wax · Hybrid rocket · Thermal decomposition · Compressive strength

Abbreviations

A_p	Cross-sectional area of combustion port
d_b	Port diameter after burnout
d_{ig}	Port diameter before ignition
G_{ox}	Oxidiser mass flux
l_f	Fuel grain length
m_b	Burnt mass of fuel
$\dot{m}_{ox}$	Oxidiser mass flow rate
t_b	Burn time
T_m	Melting temperature
ΔH_f	Heat of fusion

ΔH_d	Heat of the oxidation reaction
ρ_f	Fuel density

1 Introduction

Constant improvements are being made in hybrid rocket research to enhance performance, combustion efficiency, safety, and low-pollution exhaust combustion products [1–3]. A hybrid rocket combines the properties of both solid and liquid rocket engines. The propellant mixture of solid fuel and, most commonly, liquid oxidisers are used in the hybrid propulsion system. Hybrid rockets have the advantages of thrust modulation, restart ability, reignition, controllability, and safe handling [4]. Common propellant combinations in a hybrid propulsion system are hydroxyl-terminated polybutadiene, polyethylene (PE), and polymethyl methacrylate (PMMA) as a solid fuel and LOX or nitrous oxide (N₂O) as an oxidiser. The main drawback of these polymeric fuels in the hybrid rocket is slow fuel regression rate. The slow regression rate of polymeric fuels is due to the diffusive nature of flame in the combustion zone. The diffusive flame

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over the solid fuel limits the heat transfer to the fuel surface, resulting in slow fuel degradation to the combustion zone. The slow regression rate of solid fuel provokes low thrust performance. These intrinsic limitations are required to overcome for achieving the potential advantages of the hybrid rocket system. Several studies have been performed to increase the regression rate and ballistic performance of these polymer fuels [2, 5–10].

Paraffin-based fuels are characterised by a high regression rate, low cost, safety features, reliability, and low environmental impact. However, casting a large-scale paraffin motor is difficult because paraffin wax has poor mechanical properties [11, 12]. Karabeyoglu et al. [13] identified fast burning paraffin-based fuels that form an unstable liquid layer over the fuel surface during combustion. The regression rates of these liquefying fuels were three to four times those of the traditional polymeric fuels.

Maruyama et al. [14] studied the mechanical performance of paraffin-based fuels doped with ethylene vinyl acetate (EVA). They reported that the maximum tensile strength improved up to approximately 122% when 20 wt% EVA was added to the paraffin wax. The maximum strain increased by 280% relative to that of the control sample. At the same EVA loading in paraffin wax, the regression rate decreased by nearly 35% relative to that of pure paraffin wax. Kim et al. [15] studied the effect of PE and paraffin blended fuel in a hybrid rocket motor. The study results indicated that the regression rates of the paraffin-based fuels increased by approximately 3.9 times relative to those of pure high-density PE. Moreover, the mechanical strength of the blended fuel increased with the addition of PE.

The combustion efficiency of paraffin wax is lower than that of traditional polymeric fuels [16]. In hybrid rocket combustion, paraffin fuel droplets striped from the solid fuel surface leave the combustion chamber without undergoing complete combustion. Additives such as PE, Al, B, and CB can be used to improve the thermal stability, mechanical strength, and combustion performance of paraffin-based fuels.

In this study, paraffin-based solid fuels were manufactured using various additives such as PE, Al, B, and CB to enhance the fuels' mechanical strength and thermal properties. Blending pure paraffin wax with PE has several advantages such as a uniform material structure, which improves

the mechanical properties and combustion efficiency. Three classes of paraffin-based fuels were manufactured using the typical melt method, and mechanical strength was evaluated using compressive tests. Differential scanning calorimetry (DSC) was performed to study the melting enthalpy, exothermic reaction temperatures, and heat of combustion of these solid fuels. Finally, ballistic tests were conducted to study the regression rates of these paraffin-based fuels.

2 Materials and characterisation

The materials used for fuel fabrication in this study are listed in Table 1. Paraffin wax with a melting point of 59–66 °C, molecular weight of 394 g/mol, and density of 920 kg/m³ at 25 °C was supplied by Merck Millipore Pvt. Ltd. PE in pellet form with a melting point of 104 °C, molecular weight of 96000 g/mol, and density of 918 kg/m³ was used to strengthen the paraffin wax. Al, B, and CB of 99.5% purity and having a particle size between 6.5 and 19 µm were added to increase the energy density of the prepared solid fuels.

Three types of paraffin-based fuels were prepared using Al, B, and CB as additives. To improve the structural and thermal stability of the paraffin-based fuels, 10 wt% PE was added to the fuel formulations. A pure paraffin sample was selected as a reference formulation. Table 2 presents the composition of the prepared paraffin-based fuel formulations. The fuel samples for the mechanical and ballistic test were prepared using a similar process as described in the cited literature [10, 17]. Generally, the inclusion of CB (acts as opacifying agent) into the solid improve its absorptivity and increase

Table 1 Physical properties of as-received materials

Properties	Paraffin	Polyethylene	Boron	Aluminum	Carbon black
Density, (kg/m ³)	920	918	2370	2700	1.8–2.1
Molecular weight, (g/mol)	394	96000	10.81	26.59	12.01
Melting point, (°C)	59–66	104	2065	660	3827
Viscosity, (at 100 °C), (Ns/m ²)	6–8	–	–	–	–
Average particle size (µm)	–	–	8.1	6.5	11–19

Table 2 Fuel composition of prepared samples

Fuel formula- tion	Paraffin (%)	Polyeth- ylene (%)	Alu- minum (%)	Boron (%)	Carbon black (%)
P	100	–	–	–	–
P–PE	90	10	–	–	–
P–Al	75	10	15	–	–
P–B	75	10	–	15	–
P–CB	88.5	10	–	–	1.5

the radiation heat feedback from CB particle in the combustion zone to unburnt fuel surface. In the current study, not all the paraffin grain were blackened with CB but only the P-CB formulation was blackened. The P-CB fuel formulation was prepared as an individual fuel sample to compare its regression rate performance with other A1 and B additives. The cast paraffin-based fuel grain is depicted in Fig. 1. To assess voids or cracks, the sample was fractured at room temperature, and the fractured surface was visually inspected.

2.1 Scanning electron microscopy

The surface morphology of the paraffin-based fuel samples was analysed using a scanning electron microscopy (SEM) instrument (Coxem CX-200, South Korea). Paraffin wax is a nonconductive material. Therefore, the prepared paraffin fuel samples were coated with a thin layer of Au–Pd to impart electrical conduction through the sample surface. Before analysis through SEM, a double-sided conductive carbon tape was used to hold the sample on the stub and scanned using a beam with an acceleration voltage of 15 kV.

2.2 Mechanical performance tests

The paraffin wax behaves as a brittle material compared to conventional polymeric fuels. The brittle materials are generally weak in tension but strong in compression. In the current study, the compression tests are conducted to know the ultimate compressive strength of the paraffin-based fuel. The tests were performed using an INSTRON (3382)-25-kN universal testing machine. Cylindrical test samples with a diameter of 33 mm and length of 25 mm were fabricated according to the ASTM D575-91 standard. The tests were performed at a crosshead speed of 3 mm/min. To ensure the accuracy of the test, each sample was tested three times. Stress–strain curves that indicated large differences in the properties were neglected for compressive strength and modulus calculations.

2.3 DSC procedure

DSC was performed using an STA 449 NETZSCH instrument. A sample of weight 10 mg was placed in a hermetically sealed Al pan. Under an oxygen atmosphere, the temperature was ramped up to 800 °C at a heating rate of 10 °C/min. The melting temperatures, melting enthalpies, peak exothermic reaction temperatures, and combustion enthalpies were calculated. These calculations are presented in Sect. 3.

2.4 Ballistic tests

The ballistic performance of the paraffin-based fuels was assessed using a lab-scale ballistic test motor. Gaseous oxygen served as the oxidiser. The lab-scale hybrid motor setup comprised an oxidiser supply system, a stainless steel combustion chamber, convergent–divergent nozzle, and pyrogen ignition system. The oxidiser mass flow rate into the combustion chamber was controlled and regulated using a pressure regulator. Gaseous oxygen was supplied into an oxidizer-settling chamber of diameter 55 mm and length 33 mm. The oxidiser was injected into the combustion chamber through an axial-flow injector plate. To ensure the axial injection of the oxidiser, the injector plate was drilled with 14 holes each of diameter 1 mm. The cylindrical combustion chamber with an inner diameter of 42 mm and length of 150 mm was used to accommodate the solid fuel. The fuel grain length was maintained at 142 mm in the combustion chamber to provide an 8-mm gap between the fuel port and the nozzle entry. This post-combustion chamber allowed unburnt paraffin to react with gaseous oxygen before paraffin left the nozzle. The scheme of the lab-scale hybrid rocket experimental test facility is shown in Fig. 2. To ensure the accuracy of the test, a systematic firing test procedure was followed. A typical firing test of the P–PE fuel sample is depicted in Fig. 3.

Combustion was initiated using a pyrogen igniter loaded through the nozzle end of the motor. Combustion extinction was achieved by remotely closing the oxidiser flow supply after the burn time. Several test parameters such as the burnt fuel weight, oxidiser mass flow rate, and the average final

Fig. 1 Paraffin-based fuels, **a** cylindrical port configuration, **b** P–Al casted grain



Fig. 2 Schematic of the hybrid rocket experimental setup

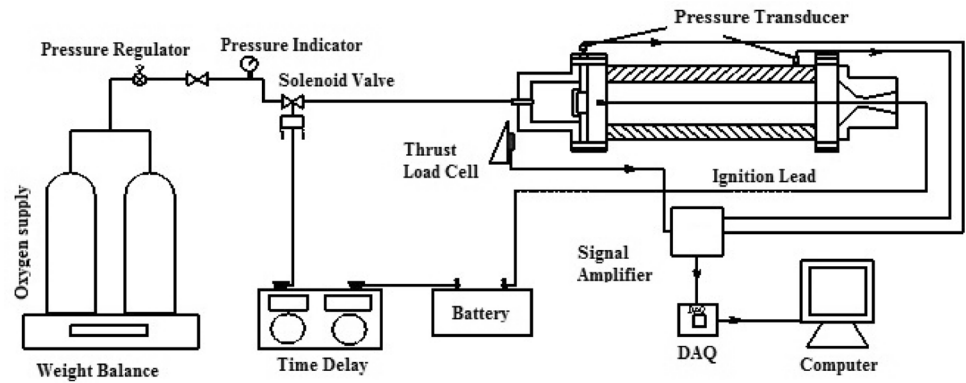


Fig. 3 Firing test of the P-PE; (inset) fuel port after firing test

diameter were recorded after each firing test. A digital pressure transducer was used to measure the chamber pressure during the motor operation. The pressure transducer was an Omega PX603 s. The operating pressure range of this sensor was 6.89 MPa with a $\pm 0.5\%$ accuracy of the full span. This sensor provided a 4–20 mA output voltage. The pressure transducer was positioned at the nozzle end of the combustion chamber (post-combustion chamber). The thrust was measured using a column load cell of 20 kg load capacity. The load cell was mounted near the oxidizer assembly of the motor. During the testing, the motor slides over the ball bearing rail and translated the force developed by the motor to the load cell. The regression rate of solid fuel was calculated using the mass loss method [1, 10, 18]. The average regression rate can be calculated from the average diameter change and burning time (t_b) using the following Eq (1):

$$\dot{r} = \frac{d_b - d_{ig}}{2t_b}, \quad (1)$$

where d_{ig} is the diameter of the fuel grain before the ignition process. In above equation, the burning time (t_b) is measured as the time between the ignition and termination of the oxidiser flow. The fuel grain diameter (d_b) is calculated after the combustion extinction. The final fuel port diameter (d_b) is measured from the initial diameter (d_{ig}), the burnt mass of fuel (m_b), length of fuel grain (l_f), and density of the fuel grain (ρ_f) through the following Eq. (2);

$$d_b = \sqrt{\left(d_{ig}^2 + \frac{m_b}{\frac{\pi}{4}\rho_f l_f}\right)}. \quad (2)$$

The oxygen mass flux can be estimated from the mass flow rate of oxygen ($\dot{m}_{ox}$) and cross-sectional area of the fuel grain (A_p) using the following Eq. (3);

$$G_{ox} = \frac{\dot{m}_{ox}}{A_p}. \quad (3)$$

The cross-sectional area of the fuel grain (A_p) is expressed as follows:

$$A_p = \frac{\pi}{4} \left(\frac{d_b + d_{ig}}{2} \right)^2. \quad (4)$$

In the above equation, the fuel port diameter before the ignition event and after the burnout is considered to calculate the combustion port area.

3 Results and discussion

3.1 SEM results

SEM analysis was performed to study the phase structures of the paraffin-based fuels. A near-uniform phase structure of the pure paraffin wax can be observed in Fig. 4a. Moreover, paraffin loaded with PE, Al, and CB exhibited a uniform structure without any phase separation. The uniform

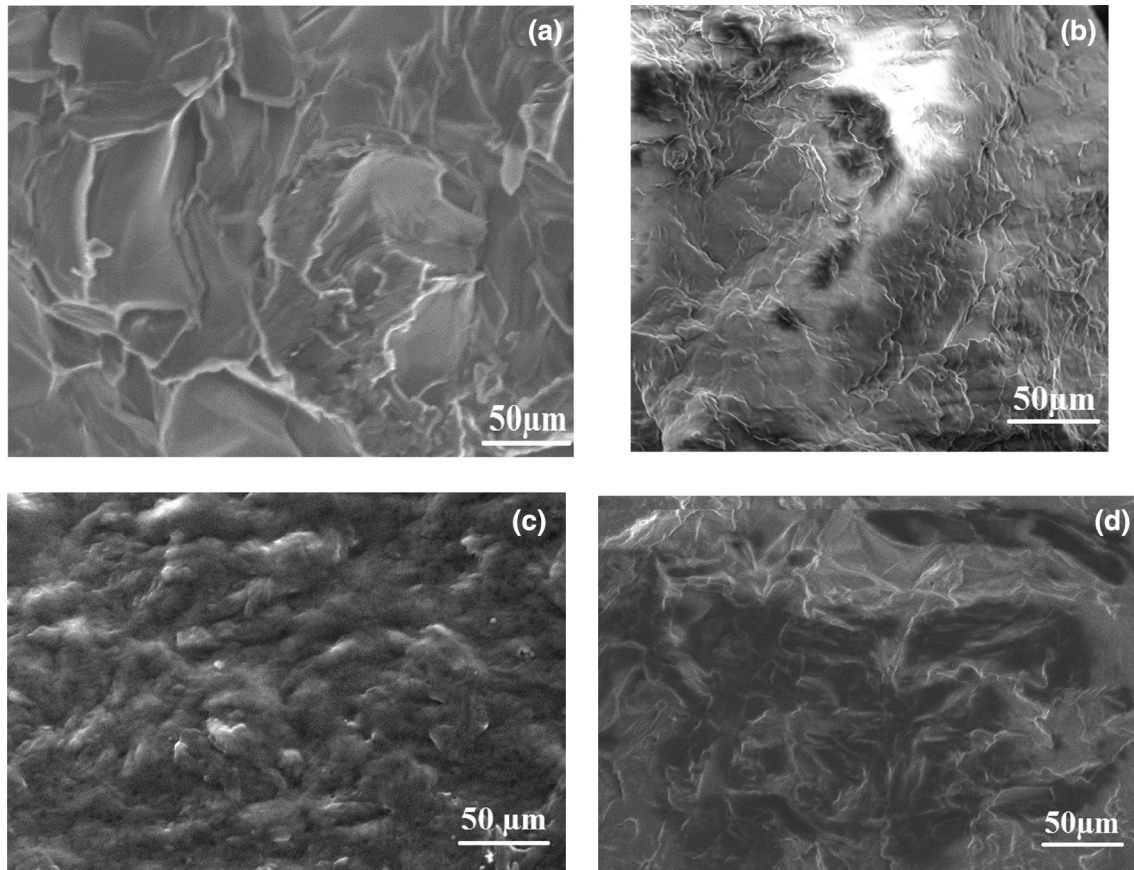


Fig. 4 SEM micrographs, **a** pure paraffin wax, **b** P-PE sample, **c** P-Al sample, **d** P-CB sample

morphology of the P-PE and P-Al fuels can be clearly seen in Fig. 4b–d. Any phase separation in the blend could adversely affect the mechanical properties of the paraffin-based fuels. PE and CB were dispersed homogeneously in the paraffin wax with proper interfacial bonding. Krupa et al. [19] explained the effect of the miscibility of PE and soft paraffin wax on phase separation.

3.2 Mechanical performance tests

Compressive tests were performed to evaluate the mechanical performance of the prepared paraffin-based solid fuels.

To ensure the accuracy and consistency of the test, each fuel formulation was tested three times; the average values of the test parameters from the statistical analysis are presented in Table 3. Figure 5 illustrates the average compressive modulus and compressive strength of the paraffin-based fuel formulations. The average compressive modulus and compressive strength of the P-PE sample increased by approximately 99.4% and 41%, respectively, compared with those of the pure paraffin sample. The improvement in mechanical properties could be due to the molecular structure and low crystallinity of the material. During solidification, paraffin wax resulted in large pieces of crystalline material [10, 11,

Table 3 Results of compressive strength and modulus of blended paraffin fuels

Sample	Average compressive strength (MPa)	Rate of increase with reference to pure paraffin (%)	Average compressive modulus (MPa)	Rate of increase with reference to pure paraffin (%)
P	3.07 ± 0.02	—	82.3 ± 6.1	—
P-PE	4.33 ± 0.19	41	164.1 ± 12.5	99.4
P-Al	4.89 ± 0.07	59.2	179.5 ± 14.2	118.1
P-B	4.73 ± 0.2	54.1	171.4 ± 10.7	108.2
P-CB	4.41 ± 0.23	43.6	169.2 ± 12.6	105.6

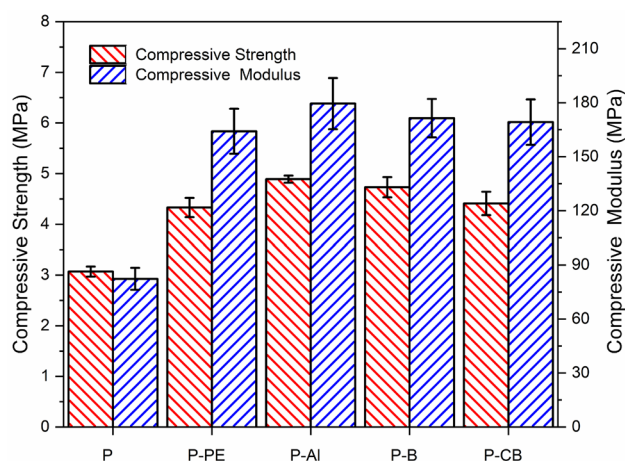


Fig. 5 Comparison of compressive strength and modulus of blended paraffin fuels

20]. The crystallinity of the wax decreased with the addition of PE. Blending paraffin wax with PE could result in a uniform mixture.

Adding 15 wt% Al to the P-PE sample improved the average compressive strength and modulus by nearly 59.2% and 118.1%, respectively, compared with those of the unmodified paraffin. Furthermore, adding 15 wt% B and 1.5 wt% CB increased the average compressive strength by approximately 54.1% and 43.6%, respectively. The addition of metallic additives reduced the chain mobility of the P-PE matrices, which resulted in a stiff and tough formulation [18]. Veale et al. [12] have performed tensile tests on a sample of paraffin fuel reinforced with Al powder and observed similar improvements in mechanical strength. Moreover, adding PE to paraffin can form strong interfacial adhesion, which can improve mechanical strength through effective stress transfer. Table 3 demonstrates that the mechanical strength of the paraffin-based fuels is sufficient to prevent vibrations, ignition shocks, and structural deformation loads during the handling and transportation of these fuels for hybrid rocket applications.

3.3 DSC results

DSC analysis was performed on the paraffin-based fuels loaded with various additives to investigate the effects of the additives on thermal decomposition. At low temperature, two endothermic peaks were observed in the DSC curve, and the peaks were determined to correspond to the solid–solid transition and melting of the paraffin wax (Fig. 6a). The pure paraffin wax sample melted at approximately 64.4 °C (T_{m1}), and PE started melting at approximately 110.4 °C (T_{m2}). The area under the endothermic peak in Fig. 6a represents the melting enthalpy of the fuel sample. The total melting enthalpy of the formulation was evaluated by adding the

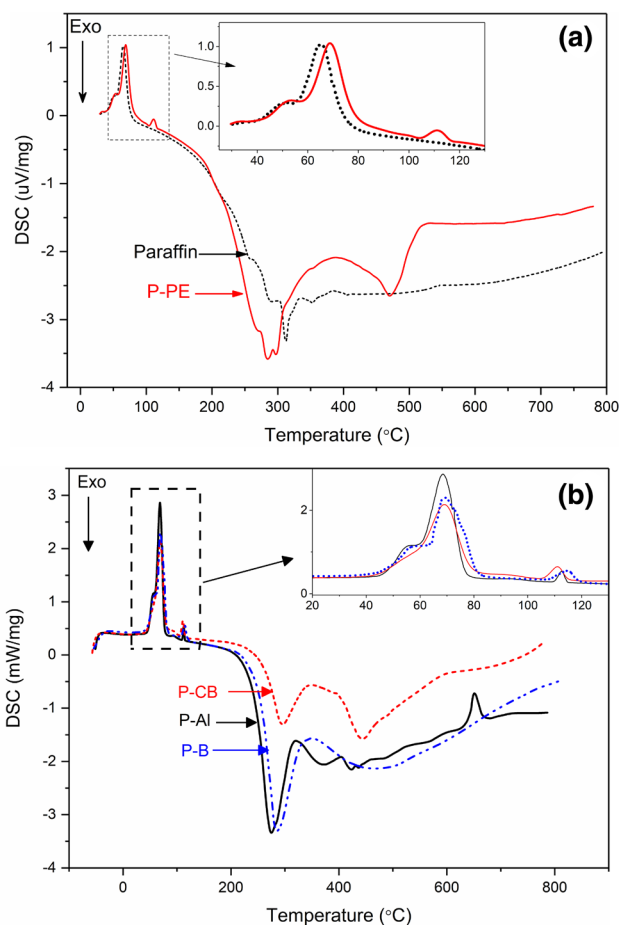


Fig. 6 DSC heating curves **a** P and P/PE samples, **b** P/PE/additives

melting enthalpies of all endothermic peaks. The total melting enthalpy of the paraffin wax was nearly 8.9 J/g for the P sample and 7.9 J/g for the PE sample.

The total melting enthalpy of the paraffin sample decreased with the addition of PE to paraffin because of the low crystallinity of the wax in the P-PE formulation. With the addition of PE to the paraffin wax, the concentration of the wax decreased in the formulation, which in turn reduced the crystallinity of the formulation. A relatively low melting enthalpy can promote the regression rate of a solid fuel [15]. However, this study revealed contradictory results regarding the melting temperatures of the paraffin-based fuels, indicating that the addition of additives raised the melting temperature (T_{m1}). The high melting temperature could hinder the entrainment process of these liquefying fuels.

Figure 6a illustrates an exothermic peak at 311.2 °C, which could be attributed to the oxidation of the paraffin wax. The DSC thermogram of the P-PE sample exhibited two exothermic peaks at 283.8 and 469.3 °C, which were determined to correspond to the oxidation of paraffin and PE, respectively. Oxidation of paraffin at a lower temperature can also be attributed to its softness (less thermally stable)

and much shorter chains length than the pure PE. The presence of an exothermic peak at 469.3 °C can also be attributed to the reticulation/cracking of polymeric chains of paraffin and PE blend. The heat released during the oxidation process was evaluated using the area under these exothermic peaks. The heat released by the paraffin wax sample was nearly 142 J/g, whereas that released by the P–PE sample was 222 J/g. The higher heat release in the P–PE sample was due to the oxidation of PE along with pure paraffin wax. The two exothermic peaks observed in the DSC thermogram of the P–PE sample confirm the oxidation of paraffin followed by the oxidation of the PE.

The effects of the Al (15 wt%), B (15 wt%), and CB (1.5 wt%) additives on the exothermic reaction were studied at a heating rate of 10 °C/min, and the results are presented in Fig. 6b. The P–Al fuel demonstrated two exothermic peaks followed by an endothermic peak. The endothermic peak at 649.6 °C was attributed to the melting of Al. As presented in Table 4, the peak exothermic temperature ($T_{\max 1}$) decreased for the additive samples compared with the pristine sample. This result indicates that the exothermic reaction started earlier

This early reaction may be due to the high radiative heat transfer from the additives to the surface of fuel, which raised the conductive heat transfer into the interior of the fuel and reduced the exothermic reaction temperature of the fuel sample. The P–Al sample released the highest amount of heat among all the tested samples. This indicates Al could be a superior additive to enhance heat feedback to solid fuel in hybrid rocket combustion. Based on these results, the heat energy released by these fuels can be predicted and the parameters from the DSC results can be useful for combustion modelling.

3.4 Ballistic tests

The regression rate performance of the liquefying fuels loaded with various additives was studied using a lab-scale ballistic evaluation motor. All tests were performed under a gaseous oxygen environment with varying oxidiser mass flux from 56.1 to 94.8 kg/m²s. The average regression rates

of all the tested fuel formulations are presented in Fig. 7. Notably, the pure paraffin fuel formulation had the highest regression rate among the paraffin-based fuels. At an oxidiser mass flux of 58.07 kg/m²s, the average regression rate of the pure paraffin sample was 1.82 mm/s. When the oxidiser mass flux increased to 83.18 kg/m²s, the regression rate was increased to 2.39 mm/s. At the same oxidiser mass flux range, the regression rate of the P–PE sample increased from 0.88 to 1.29 mm/s.

The regression rate of the P–PE blended fuel decreased considerably with the addition of the PE wax. Several studies have reported that the thermal stability of paraffin wax can be increased with the addition of PE and that polymer blending can reduce the melt viscosity of P–PE [15, 20, 21]. The regression rates of liquefying fuels depend strongly on the melt viscosity and entrainment of the thin layer of the liquid paraffin formed over the fuel surface. Pure paraffin wax has a relatively low viscosity, which drives the entrainment of paraffin droplets from the fuel surface to the combustion zone along with gasified fuel vapours. This additional component of fuel entrainment to the combustion zone

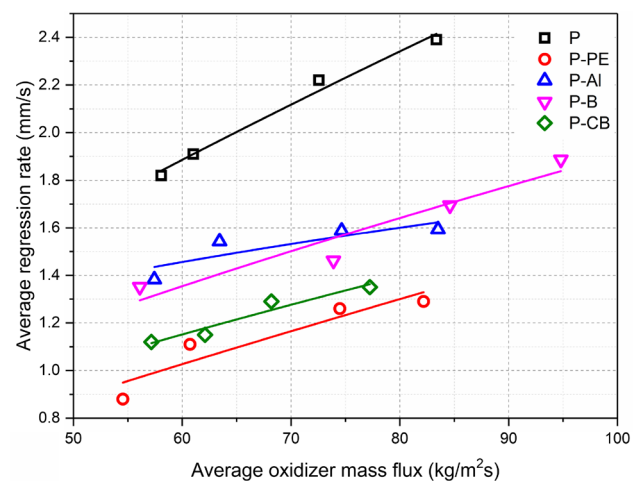


Fig. 7 Average regression rates of liquefying fuels under gaseous oxygen

Table 4 DSC results for paraffin-based fuels

Sample	Melting temperature, T_{m1} (°C)	Melting temperature, T_{m2} (°C)	Melting temperature, T_{m3} (°C)	Heat of fusion, ΔH_{f1} (J/g)	Heat of fusion, ΔH_{f2} (J/g)	Total Heat of fusion, ΔH_f (J/g)	Peak exothermic temp., $T_{\max 1}$ (°C)	Peak exothermic temp., $T_{\max 2}$ (°C)	Heat of oxidation reaction ΔH_d (J/g)
P	64.4	–	–	8.9	–	8.9	311.2	–	142
P–PE	69.4	110.4	–	7	0.91	7.9	283.8	469.3	222
P–Al	68.2	112.1	649.6	6.5	1.01	7.5	274.9	420.6	291
P–B	69.1	115.2	–	6.6	1.12	7.7	284.9	475.2	274
P–CB	69.5	110.8	–	6.3	1.16	7.5	295.4	443.6	268

improves the regression rate of paraffin wax compared with those of other traditional polymer fuels.

However, the combustion efficiency of paraffin fuels is poor and could be improved by blending pure paraffin with PE [15]. This is because a thermally stable P–PE blend can control the discharge of paraffin droplets produced from paraffin wax [21]. The Al, B, and CB additives were added to the P–PE blend to increase the energy density of the fuel. The addition of these additives could compensate for the depletion of regression rates with PE addition. As shown in Fig. 7, the Al and B metallic additives had higher regression rates than did the P–PE blended fuel. However, the regression rates were lower than that of pure paraffin wax. Adding CB to the P–PE blend improved the regression rate compared with that of P–PE, but this regression rate was lower than that of the other metallic additives. The regression rates of the Al and B metallic additives formulations increased by 57% and 53.5%, respectively, compared with that of the P–PE formulation. The increase was attributed to high heat release from the metallic additives during the combustion process. The heat feedback from the additive to the fuel surface increased the vaporisation and entrainment processes and engendered higher regression rates [22].

The regression rates of the tested paraffin-based fuels were correlated with the oxidiser mass flux through the regression rate law:

$$\dot{r}_f = aG_o^n \quad (5)$$

where the ballistic coefficients a and n were calculated from the experimental data. As shown in Table 5, the dependency of the regression rate on oxidiser mass flux was similar between the pure paraffin and P–PE fuel samples. Moreover, the dependency of the regression rates on oxidiser mass flux was similar among the fuel samples loaded with additives. The improved regression rate performance of the paraffin-based fuels could be attributed to high radiative heat transfer and high fuel surface roughness over the entire range of oxidiser mass flux.

The thrust-time traces combustion pressure–time traces observed for the paraffin-based fuel formulations are illustrated in Fig. 8. Major events of the test motor run on the thrust-time trace were as follows: (a) slight thrust level rise during the initiation of the oxidiser supply, (b) motor ignition accompanying thrust spike, and (c) motor shut off. A trivial amount of thrust level rise was observed as the oxidiser flow was initiated. When the firing voltage was applied to the ignition squib, motor ignition occurred. The sharp thrust spike during the motor ignition process was caused by the ignition squib detonation. The detonation of the ignition squib generated gases and considerable heat, thus producing a temporary spike on the thrust curve. A sharp cut-off thrust occurred when the oxidiser flow was terminated.

Table 5 Regression rate data from lab-scale ballistic tests

Fuel sample	Final grain diameter (mm)	Initial fuel mass (g)	Final fuel mass (g)	Burn Time (s)	Average oxidiser mass flux, kg/m ² s	Regression rate (mm/s)	Regression rate exponents	
							a	n
P	27.97	715	655	3.56	58.07	1.82	0.086	0.71
	28.14	714	643	3.43	61.02	1.91		
	31.41	716	620	3.7	72.56	2.22		
	31.85	720	622	3.52	83.36	2.39		
P–PE	21.59	708	682	3.75	54.54	0.88	0.036	0.81
	23.04	728	695	3.63	60.72	1.11		
	22.84	721	689	3.1	74.48	1.26		
	23.83	735	698	3.42	82.18	1.29		
P–Al	25.31	739	692	3.73	57.44	1.38	0.132	0.62
	25.49	728	680	3.4	63.43	1.54		
	27.17	751	693	3.83	74.65	1.58		
	24.24	725	684	2.9	83.51	1.59		
P–B	26.24	760	695	3.55	56.1	1.35	0.087	0.66
	25.26	772	713	3.51	73.9	1.46		
	27.19	756	685	3.6	84.6	1.69		
	27.35	758	686	3.27	94.8	1.88		
P–CB	22.24	716	687	3.24	57.17	1.12	0.075	0.66
	22.24	722	693	3.14	62.1	1.15		
	23.26	715	681	3.21	68.19	1.29		
	24.61	718	677	4.16	77.24	1.35		

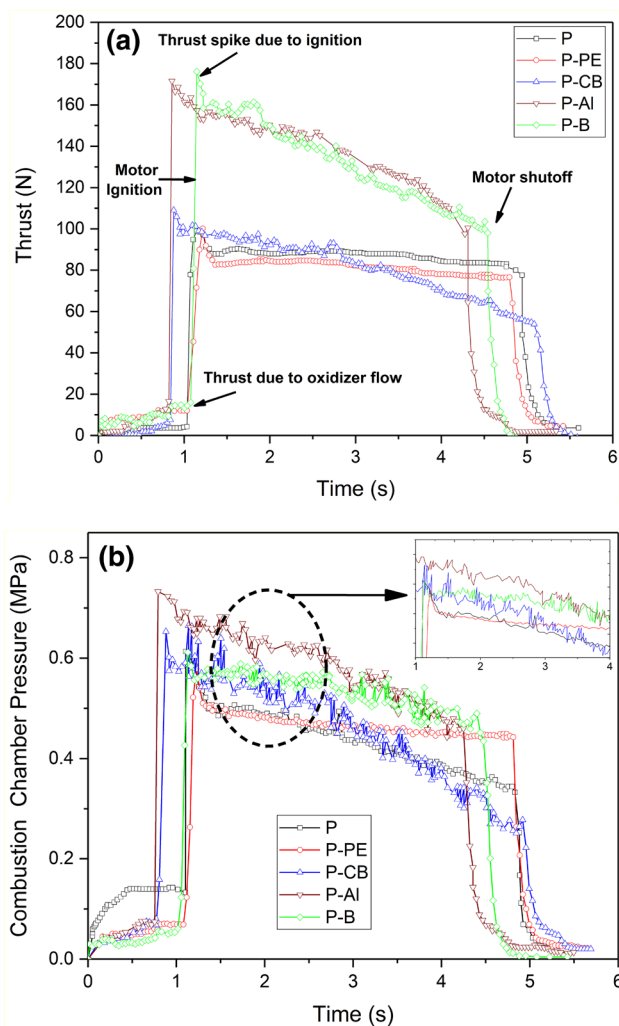


Fig. 8 **a** Thrust versus time traces of paraffin-based fuels, **b** combustion chamber pressure versus time traces of paraffin-based fuels

As illustrated in Fig. 8a, during the entire period of motor burn, the thrust profile of the pure paraffin fuel sample was smooth and exhibited stable combustion without any abnormalities. The average thrust produced by the pure paraffin wax was approximately 87.5 N at an oxidiser mass flux of 72.56 kg/m²s. At the same oxidiser mass flux range, the average thrust produced by the P-PE fuel motor was nearly 81.3 N. The addition of PE to the paraffin wax reduced the average thrust and could be attributed to the improved thermal stability of the P-PE blend. For a fuel sample, a relatively high thermal stability and viscosity could reduce the mass flow rate of the fuel to the combustion zone and result in low thrust [23]. The thrust data for the P-CB formulation revealed an average thrust of nearly 83 N at an oxidiser mass flux of 77.2 kg/m²s; in contrast, the P-B and P-Al formulations produced average thrusts of 129.5 and 135.6 N at oxidiser mass flux values of 73.9 and 63.4 kg/m²s, respectively.

Combustion chamber pressure over time traces of paraffin-based fuels is presented in Fig. 8b. It can be observed that the chamber pressure profile of all paraffin-based fuel is not constant throughout the motor burn time. The pressure shows a decreasing trend during the course of the motor run, which was due to the change in oxidizer mass flux as the fuel port diameter enlarges. Combustion profile of Paraffin and P-PE fuels are smooth and indicating stable combustion without any abnormalities compared to that of loaded with additives. Pure Paraffin fuel sample achieved an average combustion chamber pressure of approximately 0.45 MPa and 0.47 MPa was observed for the P-PE fuel sample. The addition of CB, Al and B additives into P-PE sample show a considerable amount of fluctuation in the chamber pressure. P-CB sample has shown an average chamber pressure of about 0.51 MPa, whereas the P-B and P-Al samples produced an average chamber pressure of 0.57 MPa and 0.63 MPa, respectively. The addition of metal additives into solid fuel can suppress the pressure oscillation during the combustion process [6]. During the combustion process, the metal additives ejected from the solid fuel surface can damp the combustion pressure oscillation. However, in the current study, metallic additive-based fuels have shown a contradictory trend of pressure oscillation. The cause of the noise level in chamber pressure data of additives loaded fuel is not known at this time and need further investigation.

The addition of additives to the P-PE blend increased the average thrust of the fuel formulation. In addition, the fuel formulation containing the CB additive provided a low average thrust value compare with those containing the other metallic additives. This could be attributed to carbon functioning as a heat sink that absorbs a portion of the heat of combustion in the flame zone rather than swiftly transferring the heat to the solid fuel [1]. The average thrust value of the P-Al formulation was higher than that of the P-B formulation. This could be attributed to the combustion of B particles. At a low combustion pressure, the ignition and combustion of B particles are difficult and require a high temperature and pressure to break the boron oxide layer formed over the core B particles [6].

The addition of metal additives to the P-PE blend increased the flame temperature and radiative heat flux in the combustion zone and improved the thrust. Furthermore, the average particle size of Al was determined to be smaller than those of the CB and B additives. The small particle size increased the reaction surface area, leading to enhanced fuel burning. Notably, the addition of metal additives displayed the declining nature of the thrust-time curve as compared to that of non-metallic additives. This could be attributed to nozzle throat erosion caused by metallic particles in the high-velocity exhaust gas. The combustion of metallic additives produces oxide particles, which impinge on the wall of the nozzle throat and lead to removing of nozzle wall

material. The nozzle erosion process changes the throat area of the nozzle and decreases the performance.

4 Conclusion

This study investigated the effect of various additives on the ballistic, thermal, and mechanical characteristics of liquefying paraffin fuels. The results reveal that the compressive modulus and strength of the P-PE formulation increased with the addition of PE compared with that of the pure paraffin sample. The addition of Al, B, and CB additives further improved the compressive modulus and strength of the paraffin-based fuel formulations. The DSC results show that the melting enthalpy of the paraffin-based fuels decreased with the addition of additives. Fuel formulations with low melting enthalpies can favour the entrainment phenomenon in hybrid rocket combustion. When the B, Al, and CB additives were added to the paraffin-based fuels, they accelerated the oxidation process and reduced the peak exothermic reaction temperatures. The paraffin-based fuel loaded with the Al additive exhibited a higher level of heat release during oxidation than did the samples loaded with the B and CB additives. Finally, lab-scale ballistic test results demonstrate that adding PE to the paraffin wax reduced the regression rates, whereas the Al, B, and CB additives considerably improved the regression rates of the fuel samples. The DSC results can be used to predict the combustion process of liquefying paraffin fuels. These mechanically stiff solid fuels can be a favourable choice for tactical missiles and hybrid rocket applications to sustain the thermal and pressure loads during operation.

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